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Late Cretaceous paleomagnetic results from southeastern China, and their geological implication

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Abstract

A paleomagnetic study was carried out on late Cretaceous red beds in central Jiangxi Province, southeastern China. Stepwise thermal demagnetization was used to isolate the characteristic higher temperature component (HTC) from the Jishui (D=355.7°, I=34.8°, α_{95}=6.3°) and Ganzhou (D=15.6°, I=35.6°, α_{95}=5.5°) areas, respectively. The HTC direction from the Jishui area passes Enkin's fold test at 95% confidence level, and indicates an 18.9°±5.5° counter-clockwise rotation relative to the referent pole of the stable South China Block (SCB). The HTC direction from Ganzhou passes both McFadden's fold test at 95% confidence level and reversal test. Comparing these two new poles (81.0° N, 322.2° E, A_{95}=5.8° from Jishui, and 74.4° N, 225.1° E, A_{95}=5.2° from Ganzhou) with other coeval poles reported from South China, these results demonstrate a significant difference in paleolatitude (9.9°±4.1°) between coastal and inland areas, which might be either related to dextral shearing faulting between them or inclination shallowing conducted by sedimentation and/or compaction. However, results obtained from both anisotropy of the isothermal remanence (AIR) method and corrected results of the elongation/inclination model of Tauxe and Kent [L. Tauxe, D.V. Kent, A simplified statistical modal for the geomagnetic field and the detection of shallow bias in paleomagnetic inclinations: was the ancient magnetic field dipolar? In: J.E.T. Channell, D. V. Kent, W. Lowrie, J. Meert (Eds.), Timescales of the Paleomagnetic Field, Geophys. Monogr. Am. Geophys. Union, 145 (2004) 101-115.] indicate that a distinct inclination flattening occurred in our samples.
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Keywords: South China Block; late Cretaceous; paleomagnetism; inclination shallowing

1. Introduction

The South China Block (SCB) consists of the relatively stable Yangtze Block (YZB) and the Southeastern Fold Belt (SFB), which has suffered complex tectonic evolution since the late Proterozoic (Fig. 1a). The subduction of the Pacific plate on the eastern side and collision–penetration of Indian plate on the western side have been playing important roles in the tectonic development of SCB since the late Mesozoic. The former has taken place with diverse directions, slowdown rate and steeping angle of subduction at least since the Jurassic [1–4], which resulted in a slip fault system oriented mainly in the NE–SW direction [5]. The latter has occurred since the Cretaceous/Tertiary boundary or around 55 Ma [6,7], which led to the extrusion of the Indochina block, the opening of the South China Sea and deformations in the Longmen Shan and the eastern margin of the Sichuan

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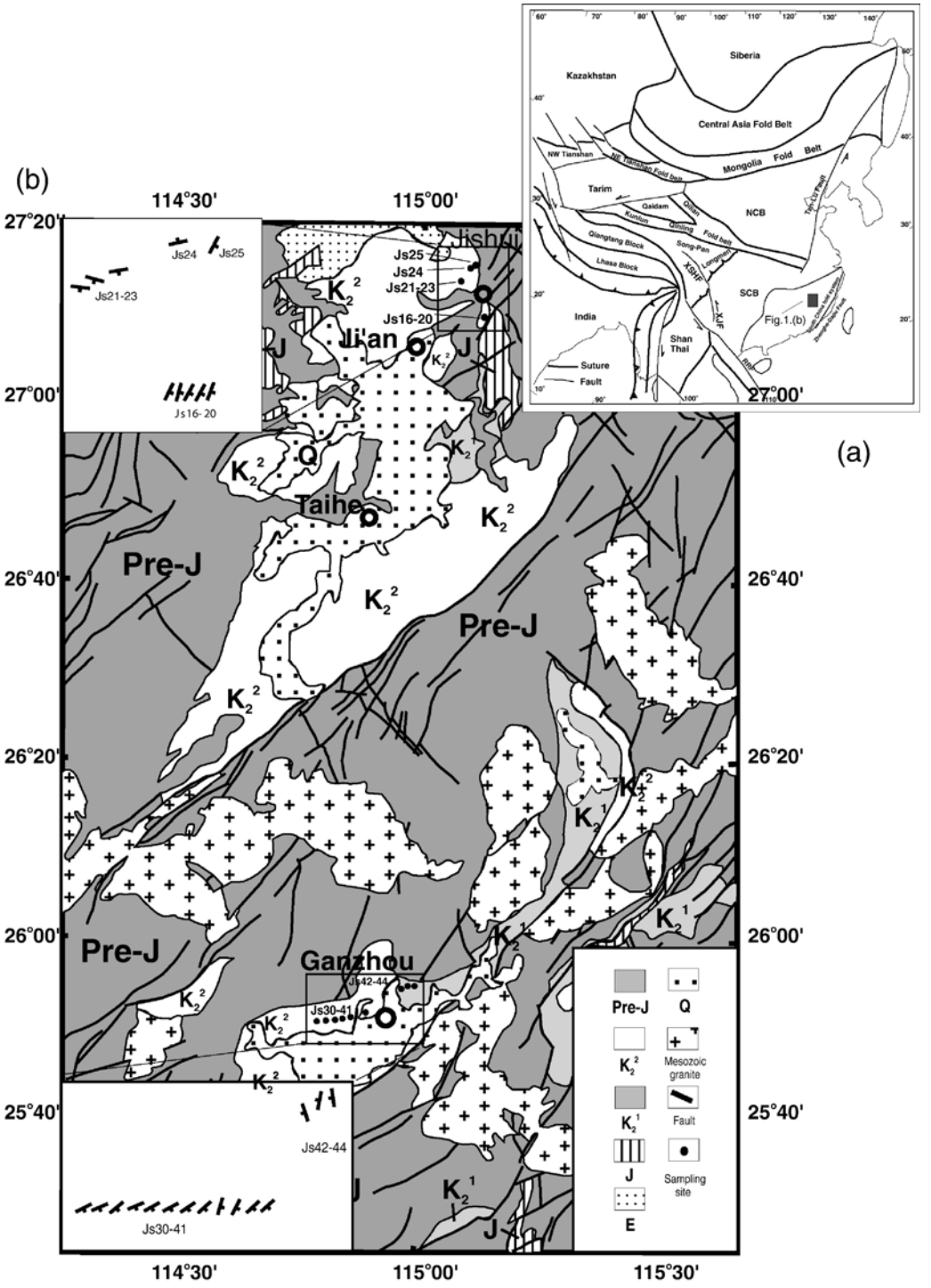


Fig. 1. (a) Simplified tectonic map of China and adjacent area (Modified from [28]). (b) Geological map of sampling areas showing the localities of late Cretaceous sites (Modified from 1:500,000 geological maps in [38]).

basin [8,9]. The SCB, therefore, is an ideal region for geologists to study the tectonic evolution of continental deformations among the Eurasia, Pacific and India plates.

The paleomagnetic technique is a useful tool for detecting interaction among different blocks. Although many paleomagnetic studies have been carried out on the

western edges and coastal regions of the SCB [9–25], data from the inland part are still scarce. A general consensus is that the collision between the North China Block (NCB) and SCB persisted from the late Permian to the early-middle Jurassic [12], and the major Chinese blocks collided with the Eurasian continent, forming a mosaic block since the Cretaceous [12,15,26–28]. However, the tectonic evolution of SCB is still controversial. Morinaga and Liu [15] suggest that SCB ranging from Sichuan Province to the coastal provinces has behaved as a stable block since the Cretaceous, because late Cretaceous paleomagnetic poles from Zhejiang, Fujian and Guangdong provinces are coherent to coeval poles of Sichuan Province and Eurasia. However, we note that the Southeastern Fold belt has experienced complex tectonic evolutions, where the blocks were limited by a series of slip faults, e.g., Xu [5] suggested that the shear systems of eastern Asia kept on acting after the Cretaceous. Gilder [29] also suggests that the Tan–Lu fault was reactivated during the late Cretaceous to Cenozoic, with right lateral strike–slip and normal faulting under the influence of the India–Asia collision. Therefore, it is necessary to determine whether the whole SCB has been stable since the Cretaceous.

Most of Cretaceous paleomagnetic results in the SCB were obtained from red beds that were thought to be good paleomagnetic research objects because of stable remanence. However, the reliability of these results has been oppugned recently, since remarkable inclination shallowing was reported in Cretaceous and Tertiary red beds in Central Asia and other areas [30–36]. The shallow inclination implies that a large continent shortening (>1000 km) occurred in the central Asian fold belts since the late Cretaceous, which can't be confirmed by geological observations. The large discrepancies in inclination cannot be evidenced by the poorly constrained age of red beds, syn-sedimentary compaction, non-dipole field or regional geomagnetic field anomaly, and poorly constrained APWP for Eurasia [30–34]. Besides, Gilder et al. [30] suggest that the inclination shallowing was related to the faster sedimentation rates conducted by mountain building that occurred in Central Asia. Because of the lower energy environment, Cretaceous red beds in North and South China may record reliable paleolatitudes. Recently, late Cretaceous paleomagnetic studies of red beds in Yichang [35] and Hengyang [36] of internal SCB have shown evidence of obviously shallowing inclinations. Therefore, it requires much more research work to detect whether the inclination shallowing is prevalent in the red beds of SCB.

Based on new late Cretaceous paleomagnetic results from Jiangxi Province, the internal part of SCB, and its Anisotropy of Magnetic Susceptibility (AMS), Anisotropy of Isothermal Remanence (AIR) characters, and corrected results using Elongation/Inclination (E/I) model of Tauxe and Kent [37], we discuss the problem of inclination shallowing, stability and interaction among the blocks in South China since the late Cretaceous.

2. Geological setting

The sampling sections are in the Ji'an-Taihe and Ganzhou basins of Jiangxi Province, 300–400 km from the coast (Fig. 1). The strata are red siltstone and sandstone that uncomfortably overly middle Jurassic sediments. In the Ganzhou basin, the strata consist of Ganzhou and Nanxiong formations, whereas the main outcropped strata compose the Nanxiong formation in the Ji'an-Taihe basin. The Ganzhou formation is a lower part of the late Cretaceous, which mainly consists of purplish-red, brick-red fine to coarse sandstone and siltstone interbedded with gray or red mudstone and siltstone. Fossils identified include vertebrates (*Conicodontosaurus kanhsienensis*) and plants (*Onychiopsis* cf. *psilotoides*, *Ruffoldia* sp., *Coniopteris* cf. *onychioides*, *Zamites* sp., *Brachyphyllum*. sp., *Pagiophyllum* sp., *Elatocladus* sp., *Cephalotaxus* sp., *Cercidiphyllum* sp., *Quercus* sp., and *Sabalites mortana*), indicating the early stage of the late Cretaceous. The Nanxiong formation is the upper part of the late Cretaceous, and mainly consists of purplish-red, brick-red conglomerate and coarse-grained sandstone in the lower and middle parts, and brick-red sandstone, siltstone and mudstone in the upper part. Fossils identified include dinosaurs eggs (*Oölithes spheroids*, *Oölithes rugustus* and *O. elongatus*), Ostracoda (*Cristocypridea*) and gastropods (*Truncatella maxima* and *Rubeyella carinate*), indicating the late stage of the late Cretaceous [38].

We sampled ten sites (95 cores) and nine representative oriented block samples from the late Cretaceous Nanxiong formation along a village road in the southeast and north of Jishui county in the Ji'an-Taihe basin, and 15 sites (154 cores) from the Nanxiong and Ganzhou formations along a highway, and the Ganzhou formation around Shangtianxin Village northwest of Ganzhou City in the Ganzhou Basin (Fig. 1b). The age of the Ganzhou formation was constrained by a K–Ar whole rock age of 85.9 Ma (unpublished 1:50,000 geological map) from an interbedded basaltic layer in the lower part of the formation near Shangtianxin Village. The sampled strata in Jishui County and the Ganzhou City form a small anticline and a synclinal fold, respectively, which favor fold testing. All cores were collected using a portable gasoline-powered drill and oriented with a magnetic

compass. Present geomagnetic declinations were computed (−3° at Jishui county and −2.7° at Ganzhou city) using the IGRF [39].

3. Paleomagnetic findings

All measurements were carried out at the paleomagnetic laboratory at the Institute of Geomechanics in Beijing. Anisotropy of magnetic susceptibility (AMS) and pyromagnetic process were measured with a KLY-3s Kappabridge. Stepwise thermal demagnetization was carried out up to 690 °C with an ASC TD-48 oven. The natural remanent magnetization (NRM) was measured using a 2G-755 cryogenic magnetometer installed inside a set of large Helmholtz coils. The bulk magnetic susceptibility of samples was measured with a Bartington

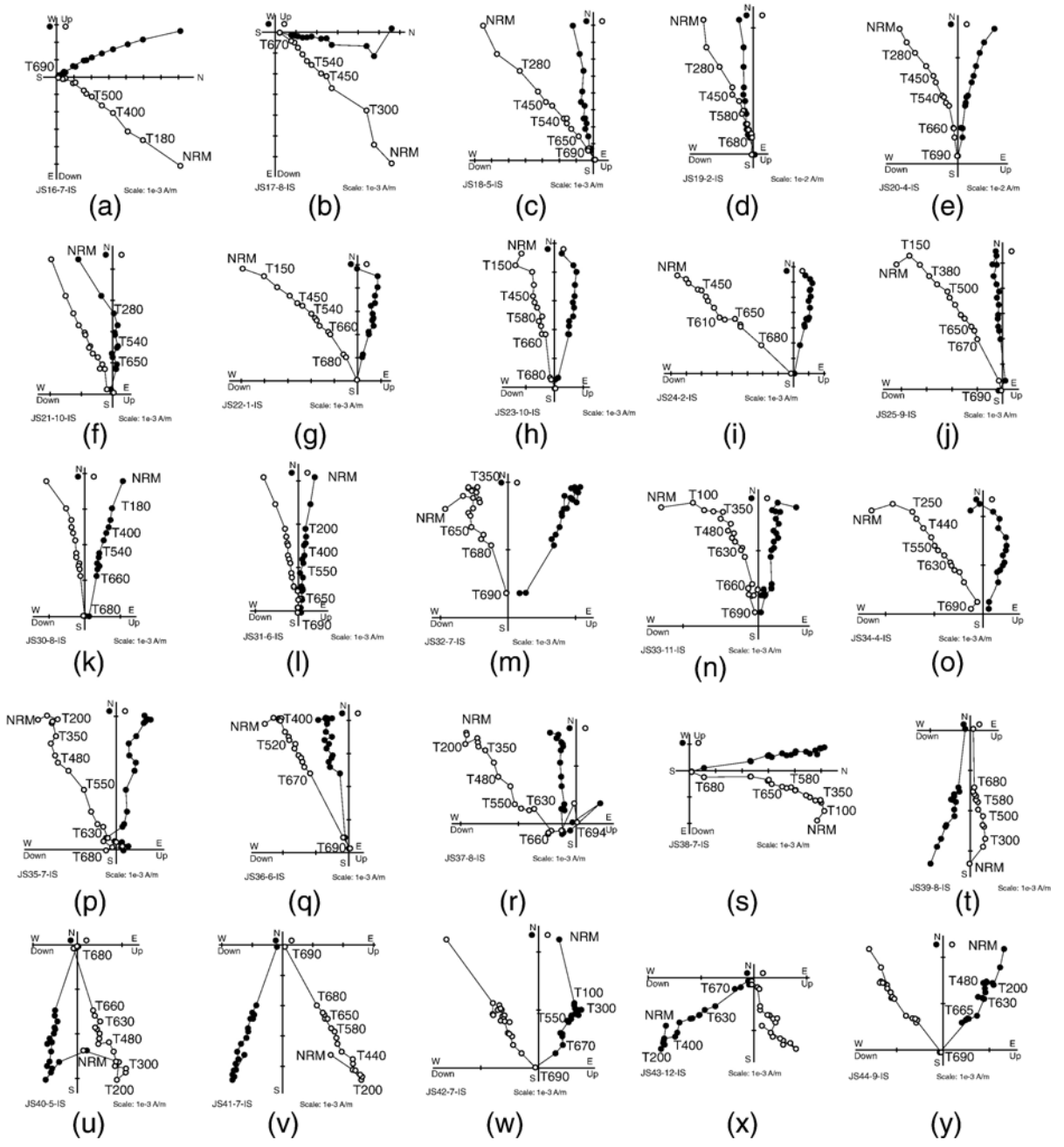


Fig. 2. Representative Zijderveld diagrams of thermal demagnetizations of NRM from Late Cretaceous samples (in-situ coordinates).

magnetic susceptibility meter at each demagnetization step, in order to monitor magnetic mineral transformations during the progressive heating. The magnetic directions were analyzed by principal component analysis [40], and the site-mean direction was calculated using Fisherian statistics [41]. To identify mineralogical carriers, 11 sister samples were selected for acquisition of isothermal remanent magnetization (IRM) and demagnetization of composite IRMs applied with three different fields along three orthogonal axes (2.2 T, 0.4 T and 0.12 T in *Z*, *Y* and *X* axes, respectively) as described by Lowrie [42]. Fifty-five samples were drilled from 9 oriented blocks in the direction perpendicular to bedding, and a remanence anisotropy test for inclination shallowing was performed with the process described by Hodych and Buchuan [43].

After removing a weak viscous magnetization between 120 and 300 °C, a single higher temperature component (HTC) was isolated during thermal demagnetization for samples from the Jishui and Ganzhou areas (Fig. 2). The high unblocking temperature (680–690 °C) implies the presence of hematite, which is also confirmed by IRM acquisition and subsequent thermal demagnetization (Fig. 3). The IRM acquisition curve shows a rapid increase below a field of 800 mT, but does not reach saturation even at 2.2 T (Fig. 3a). These results suggest that hematite is the dominant carrier of magnetic minerals in the late Cretaceous red beds of the study areas.

The site-mean HTC direction of ten sites from the Jishui area yields a north downward direction both in stratigraphic coordinate (Ds/Is=355.7°/34.8°, k=59.2, α_{95}=6.3°) and *in situ* coordinate (Dg/Ig=350.1°/27.0°, k=34.1, α_{95}=8.4°) (Table 1, Fig. 4a, b), which passes Enkin's fold test [44]. At the same time, the site-mean HTC of fourteen sites in the Ganzhou area yields an obviously different direction to that of the Jishui area in both stratigraphic coordinate (Ds/Is=15.6°/35.6°, k=53.0, α_{95}=5.5°) and geographic coordinate (Dg/Ig=10.7°/17.9°, k=23.4, α_{95}=8.4°) (Fig. 4c, d). Because only two out of eight samples from site Js37 show a stable component, and the others show erratic directions at the higher temperature steps that were not able to separate an HTC (Fig. 2r), we omit this site for further consideration. The site-mean HTC of 14 sites passes McFadden's fold test at 95% confidence level [45]. Besides, four site-mean directions of reversal polarity from the Ganzhou area are antipodal to the normal polarity directions, giving a positive reversal test with classification C [46]. Thus, we consider that both HTC directions from the late Cretaceous red beds of the Jishui and Ganzhou areas are of primary origin, acquired during the sedimentation.

4. Results of magnetic anisotropy

We performed AMS measurements on 161 samples from two sampled areas (Fig. 5) with an AGICO KLY-3s Kappabridge. The anisotropy degrees of the samples are generally high, and the anisotropy shapes are oblate forms. Directions of principal maximum ellipsoid axes (k1) in geographic coordinates are NW–SE in both

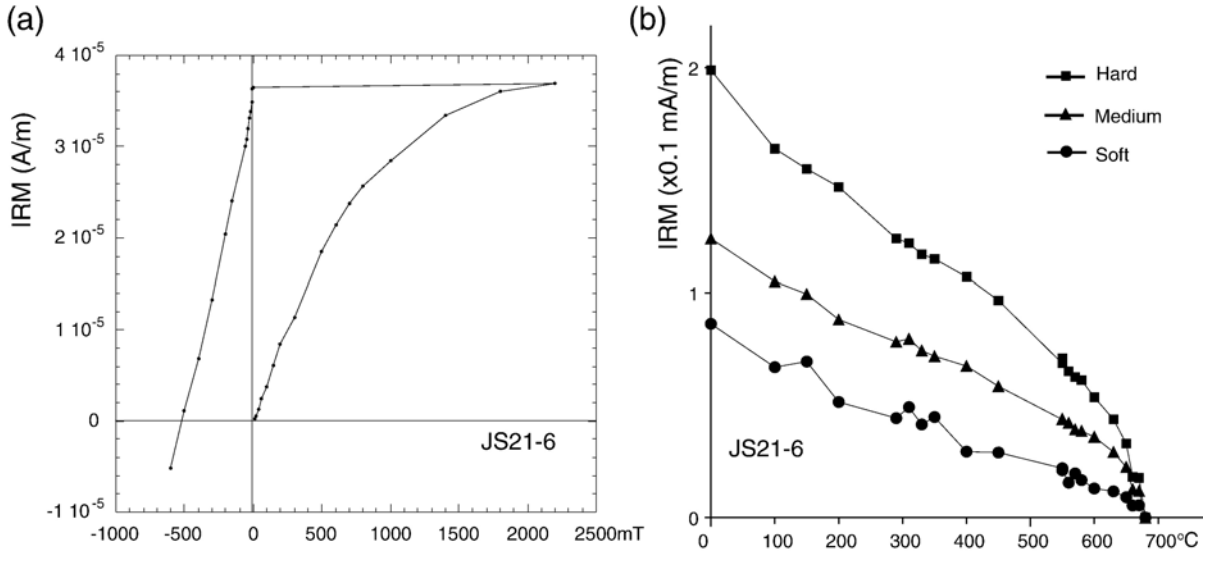


Fig. 3. (a) Isothermal remanent magnetization acquisition and opposite field demagnetization curves of red bed sample. (b) Three-component IRM thermal demagnetization curves of red bed sample.

Table 1
Paleomagnetic results from late Cretaceous red beds in Jiangxi province, South China

Site	Slat	Slong	Strike	Dip	*N*/*n*	Dg	Ig	Ds	Is	*k*	α_{95}
Jishui area											
Js16	27.1	115.1	23	26	9/9	346.9	19.2	358.1	32.7	23.2	10.9
Js17	27.1	115.1	12	16	11/11	356.2	26.9	4.7	30.2	35.7	7.7
Js18	27.1	115.1	26	33	7/8	344.5	19.3	0.1	38.2	27.7	11.7
Js19	27.1	115.1	17	26	3/3	345.0	7.8	350.8	20.6	29.7	23.1
Js20	27.1	115.1	17	26	6/7	352.1	16.5	1.7	25.6	35.8	11.3
Js21	27.2	115.1	94	14	10/12	359.4	34.5	358.2	48.4	15.5	12.7
Js22	27.2	115.1	103	9	9/11	355.8	38.1	353.2	46.6	43.6	7.9
Js23	27.2	115.1	77	7	11/11	353.4	27.2	353.8	34.1	35.5	7.8
Js24	27.2	115.1	254	20	12/12	336.0	53.1	338.0	33.2	26.6	8.6
Js25	27.2	115.1	27	17	12/12	349.8	27.1	358.8	36.2	49.1	6.3
Mean					10	350.1	27.0	355.7	34.8	59.2	6.3
Ganzhou area											
Js30	25.9	114.9	60	35	11/12	10.3	8.7	20.1	35.1	26.7	9.5
Js31	25.9	114.9	64	35	9/9	7.6	12.9	19.1	40.5	33.5	9.0
Js32	25.9	114.9	47	15	8/10	11.4	11.8	15	20.3	45.7	8.3
Js33	25.9	114.9	64	20	10/12	8.5	16.8	14.1	32.8	25.1	9.8
Js34	25.9	114.9	58	25	7/11	352.1	23.7	0.4	45.9	13.6	17.0
Js35	25.9	114.9	59	31	4/11	5.6	17.7	17.6	41.1	16.9	19.2
Js36	25.9	114.9	53	40	5/6	353.8	4.7	3.1	37.9	17.5	16.5
Js38	25.9	114.9	13	25	10/12	350.7	20.5	1.7	28.0	39.1	7.8
Js39	25.9	114.9	24	34	4/8	192.2	−15.2	203	−19.3	69.3	11.1
Js40	25.9	114.9	44	26	8/8	184.0	−15.4	193.2	−30.8	27.6	10.7
Js41	25.9	114.9	45	29	7/7	187.0	−22.9	201.9	−38.1	31.1	11.0
Js42	25.9	115.0	157	25	10/10	30.7	23.6	19.7	42.3	58.4	6.4
Js43	25.9	115.0	189	30	12/15	233.3	−24.3	216.9	−42.3	17.7	10.6
Js44	25.9	115.0	168	30	9/12	28.9	22.9	13.2	39.6	9.7	16.4
Mean					14	10.7	17.9	15.6	35.6	53.0	5.5

Slat: latitude of site; Slong: longitude of site; *N*/*n*: number of samples used to calculate mean/measured; Dg, Ig, Ds, Is: declination and inclination in geographic and stratigraphic coordinates, respectively; *k*: precision parameter; α_{95}: half angle of cone of 95% confidence about the mean direction in stratigraphic coordinate. The HTC direction from Jishui area passes Enkin fold test [44]. The HTC direction from Ganzhou area passes Mcfadden fold test [45] with $\xi1=5.246$ and $\xi2=4.700$ in geographic coordinate, $\xi1=2.068$ and $\xi2=1.083$ in stratigraphic coordinate, critical value of the test statistic $\xi=4.358$ at 95% confidence level. Moreover, this direction passes reversal test [46] with classification C.

sampling areas, with the minimum axis (*k*3) being normal to the plane defined with maximum and intermediate axes (*k*2). Directions of fold axes in the Jishui and Ganzhou areas are NE and almost NS, respectively. This indicates no relationship between magnetic fabric and the directions of local fold axes. When the magnetic fabric of hematite-bearing sandstones was affected by compressive strain corresponding to layer parallel shortening (LPS) or folding, the direction of *k*1 is usually perpendicular to that of LPS or parallel to the fold axis [47]. These cases are not presented in our results, suggesting that the red beds in our sampling areas have not experienced significant strain due to tectonic stress.

To quantitatively evaluate the possibility of inclination shallowing caused by deposition and/or compaction, we measured the anisotropy of isothermal remanence (AIR) of the samples drilled from blocks oriented in the normal direction of bedding. Because of the presence of a hematite carrier in red beds, the process described by Hodych and Buchuan [43] was employed. Fifteen samples selected from one or two samples of each block were thermally demagnetized to obtain I_{obs}. The IRM parallel (IRMx) and perpendicular (IRMz) to the bedding plane in the successive fields were measured in 40 samples, and then thermal demagnetizations were performed. The results are listed in Table 2. The value of IRMz is slightly lower than that of IRMx when the applied fields are between 200 mT and 800 mT (Fig. 6). At the same time, one or two samples from each block were collected to process the thermo-magnetic test with a KLY-3s Kappabridge in order to test magnetic mineral transformations during heating. The results indicate that few distinct chemical changes of the magnetic minerals took place during heating, and the value of IRMz/IRMx fitting after heating is reliable (Fig. 7). The mean IRMz/IRMx of thermal demagnetization between 600 and 680 °C is

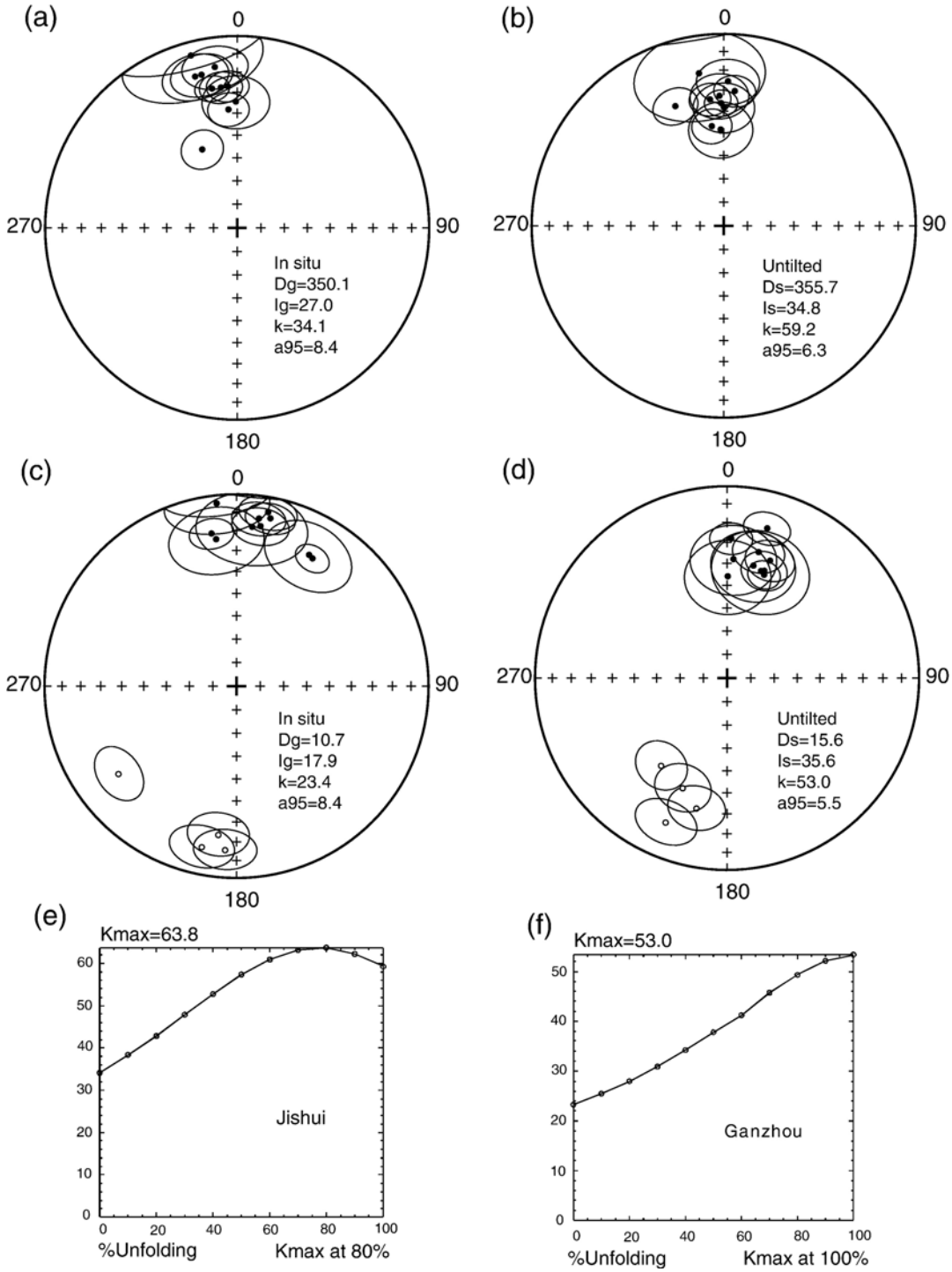


Fig. 4. Equal-area stereographic projection of site-mean ChRM from Jishui (a and b), and Ganzhou (c and d) areas in geographic and stratigraphic coordinates, respectively. (e) Progressive unfolding of the mean direction reveals a maximum concentration at 80% unfolding, in which the direction (D/I=354.3°/33.4°, α_{95}=6.0°) is statistically indistinguishable from that at 100% unfolding. (f) Progressive unfolding of the mean direction from Ganzhou reveals a maximum concentration at 100% unfolding.

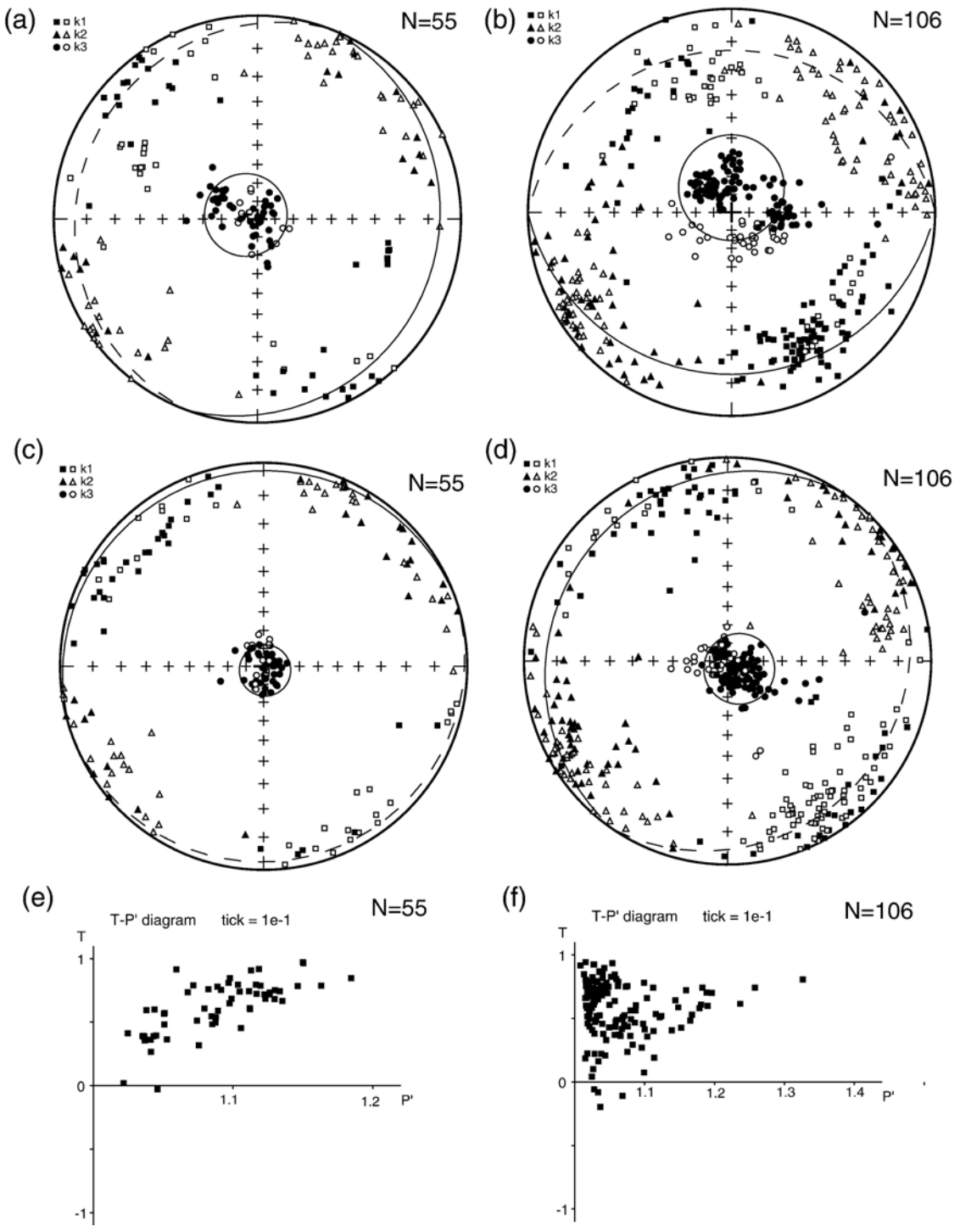


Fig. 5. Stereographic projection of AMS data in geographic (a, d) and stratigraphic coordinates (b, e) of sample from Jishui and Ganzhou areas, respectively. Plots of shape parameter (T) versus anisotropy degree (P') of samples from Jishui (c) and Ganzhou areas (f), respectively. $k1$ = maximum, $k2$ = intermediate, $k3$ = minimum). N is number of samples.

0.8222, which indicates a compaction of about 18%. Although the fitting value of IRMz/IRMx (0.8588) acquired between 200 and 800 mT is close to that defined by thermal demagnetization between 600 and 680 °C, little difference will be found when these two values are used to calculate inclination shallowing. Because

Table 2
Anisotropy of isothermal remanent magnetization for late Cretaceous red beds at Jishui area

ID	*N*/*n*	I_{obs}	IF	IRMz/IRMx (acquired between 200 and 800 mT)	IRMz/IRMx (acquired above 600 °C)
H16-1	2/9	18.6	22.3	0.8768	0.8400
H16-2	2/4	13.4	16.2	0.8378	0.7971
H16-3	2/4	20.5	24.5	0.8702	0.8059
H16-4	1/3	24.6	29.1	0.8523	0.8398
H16-5	2/5	36.0	41.5	0.8780	0.8437
H16-6	2/7	20.7	24.7	0.8398	0.7748
H16-7	1/1	43.2	48.8	0.8801	0.8301
H16-8	1/3	15.7	18.9	0.8906	0.8495
H24-1	2/4	64.9	68.9	0.8041	0.8193
Mean				0.8588	0.8222

N/*n*: number of samples suffered from the measurement of thermal demagnetization and AIR, respectively; I_{obs}: mean inclination defined from the oriented block samples.

hematite is the main magnetic carrier, the mean IRMz/IRMx from higher temperature demagnetizations (600, 620, 640, 660, 680 °C) is considered. These results are similar to that obtained from late Cretaceous red beds in the Hengyang basin of South China (IRMz/IRMx between 600 and 680 °C is 0.8828) [36]. Substituting the ratios for $\tan(I_{obs})/\tan(I_F)$=IRMz/IRMx, the corrected inclination of Jishui is about 40.2°, which demonstrates the presence of inclination flattening. Thus, we consider that depositional compaction conduced obvious inclination shallowing for the late Cretaceous red bed in Jiangxi Province, Southeast China.

5. Discussion

Although the ChRMs from the Jishui and Ganzhou areas passed the positive fold test, the folding time was not clear due to the lack of late Cenozoic strata in these areas. Local geological research reveals that folds in Jishui and Ganzhou have different ages and mechanisms. The Jishui area is located on the eastern margin of the Ji'an-Taihe basin, where a set of faults is present together with a parallel anticline and syncline in the basin. Although no younger stratum outcrops around Jishui County, except for Holocene, we found one anticline and one syncline included in the Paleogene stratum (Xinyu formation) around Xingan County situated to the north of Jishui County. The axes of these two folds are parallel to the folds in Jishui, which is overlaid by a Neogene stratum (Linjiang formation). These folds were formed in the same shearing process as the Ganjiang Fold Zone (GFZ) before the Neogene [48]. The ChRM was therefore probably acquired during the late Cretaceous.

The Ganzhou basin is a narrow basin bounded by an NE-directed normal fault on the eastern side, in which Cretaceous strata is unconformably overlain by a middle to late Neogene (Dalongli) formation. Deng et al. [48] suggested that the Ganzhou basin had suffered an extension process during the Cretaceous to Paleogene, followed by a compression process since the Neogene. The folds might have formed during this transfer process in this basin. The Neogene strata unconformably overlaying these folds indicate that the time of the folding was pre-Neogene. Taking the four successive N–R–N–R polarity sequences revealed in the Ganzhou section, and positive reversal and fold tests into account, we consider that the ChRM of the Ganzhou area was likely acquired in the late Cretaceous.

5.1. Cretaceous paleomagnetic results in the SCB

Twenty-three Cretaceous poles from the SCB were investigated in the global paleomagnetic database of Pisarevsky and McElhinny [49], which included data up to 2004. One pole obtained from Upper Cretaceous (80 Ma) dikes in Hong Kong is also included [24]. Although many paleomagnetic results have been derived from the western part of SCB, several authors suggested that the Chuan-Dian areas of the western Yangtze, located between the Red River Fault (RRF) and the Xian-Shui-He Fault Zone (XFZ) underwent different clockwise rotations relative to stable SCB and southeastward displacement after the Cretaceous [9,22,23,50]. The late Cretaceous pole of Enkin et al. [20] was obtained from the Feixiangguan and Guanyin sections of the Sichuan basin, in which only the results from the Feixiangguan section (N=7 sites) were corrected by 8° of local rotation, whereas the results from the Guanyin sections (N=9 sites) were obtained in an autochthonous terrain. The mean pole is therefore considered as one of representative poles of the western stable part of SCB.

Several reliable results have been reported recently from coastal provinces in the eastern part of SCB

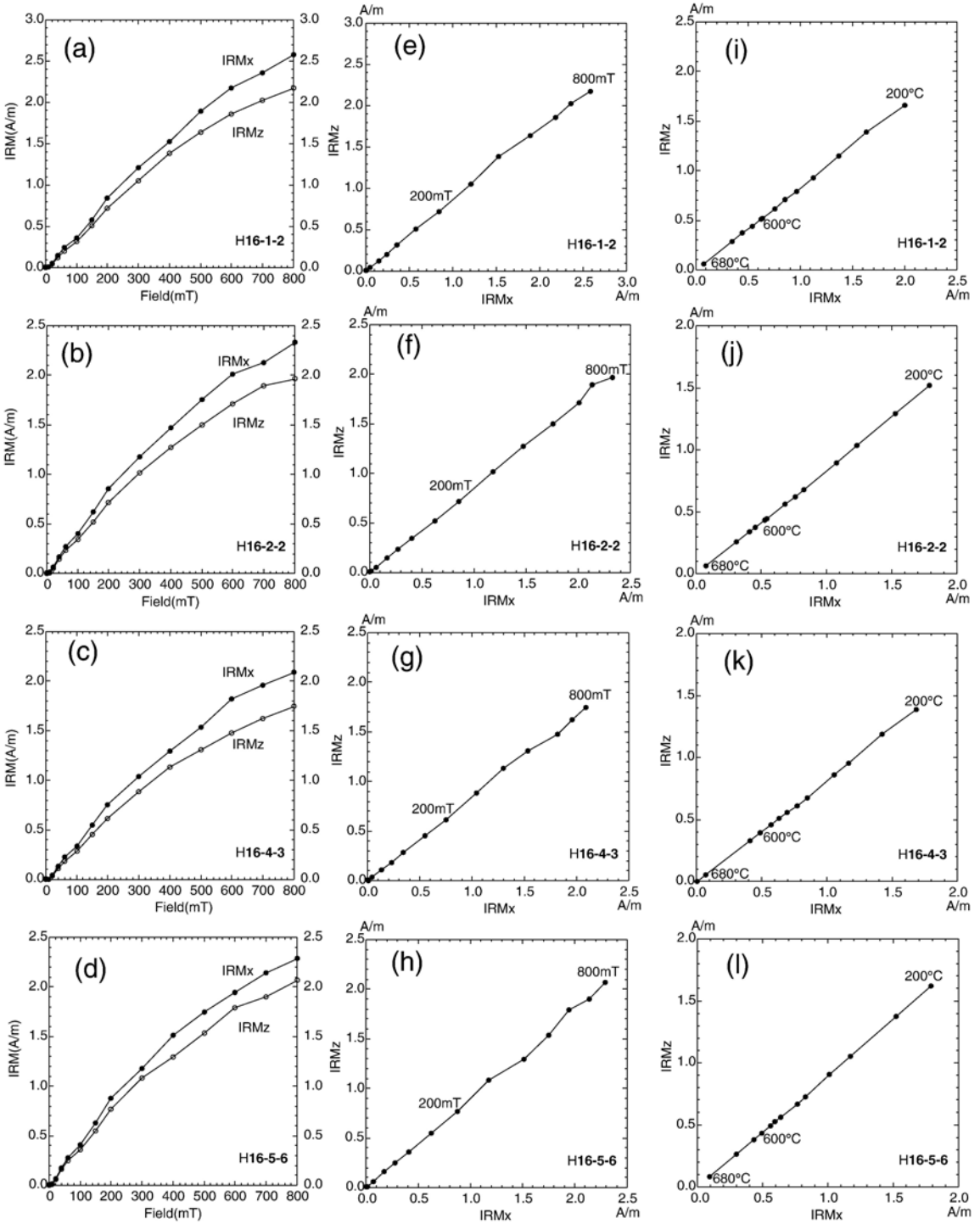


Fig. 6. Plots of IRMx (parallel to bedding) and IRMz (perpendicular to bedding) acquisitions produced by applying magnetic field at 45° to bedding as function of increasing field (a–d). The slope (IRMz/IRMx) of the least-squares-fit for data points between 200 mT and 800 mT (e–h) is used to estimate the magnetic anisotropy of hematite. The slope of the thermal demagnetization of IRMz and IRMx between 600° and 680° (i–l) provides a ratio that is used to calculate inclination shallowing of red beds samples.

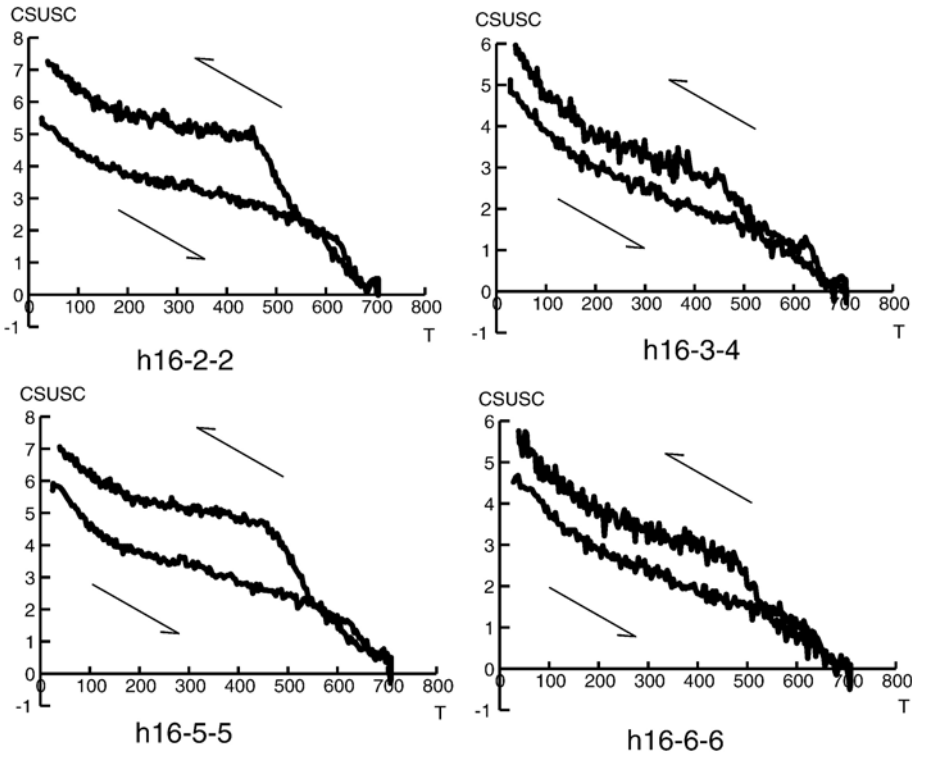


Fig. 7. Temperature dependence of magnetic susceptibility when heated from room temperature to 700 °C and subsequently cooled. There is no such phase altering to hematite in red bed sample.

(Table 3). Morinaga and Liu [15] carried out a detailed study on Cretaceous red beds from three coastal provinces. They suggested that the Cretaceous poles of Zhejiang, Fujian and Guangdong are close to the referenced poles of the Sichuan basin and Eurasia. In consequence, the SCB ranging from Sichuan to coastal

Table 3
Late Cretaceous poles from South China

Area	Location on Fig. 8	Slat.	Slong.	*N*	Plat.	Plong.	A_{95}	Reference
Western edge of SCB								
Sichuan	1	30	102.9	16	72.8	241.1	5	[20]
Eastern edge of SCB								
Nanjing, Jiangsu	2	32	119	10	76.3	172.6	10.3	[14]
Zhejiang	3	29.0	119.7	10	83.2	247.9	10.6	C in [15]
Zhejiang	4	28.4	119.9	9	77.1	199.4	7.0	D in [15]
Fujian	5	26.4	117.8	8	76.3	197.1	12.3	E in [15]
Fujian	6	25.7	116.8	6	81.4	208.7	10.5	F in [15]
Fujian	7	25.1	116.4	5	85.5	167.8	9.9	G in [15]
Guangdong	8	24.1	115.8	7	81.9	215.6	6.1	H in [15]
Anhui	9	30.8	118.2	4	83.8	200.3	14.6	[29]
HongKong	10	22.2	114.1	12	69.3	211.2	8.9	[24]
Central SCB								
Hubei	11	30.7	111.7	18	71.7	254.1	5.6	[35]
Hunan	12	26.9	112.6	26	71.9	236.3	4.7	[36]
Jishui, Jiangxi	13	27.2	115.1	10	81.0	322.2	5.8	This study
Ganzhou, Jiangxi	14	25.9	114.9	14	74.4	225.1	5.2	This study
75 Ma conference pole	15				81.3	188.6	7.2	[52]

Slat (Plat): latitude of site (pole); Slong. (Plong.): longitude of site (pole); *N*: number of sites.

provinces behaved as a stable block. However, the Southeastern Fold belt experienced complex tectonic evolutions: e.g. strike–slip faulting system related to the subduction of the West Pacific plate, where the blocks were limited by the strike–slip faults. Sediments were mostly controlled by faulting in a series of small red bed basins. Thus, taking a single basin as an example, local rotation might have occurred throughout the basin. The paleomagnetic results of six late Cretaceous poles (C, D, E, F, G, H reported by Morinaga and Liu [15]) are consistent with data from Anhui Province (Table 3) [29]. These findings indicate that the coastal provinces may belong to a stable block, and have not experienced remarkable tectonic deformation since the late Cretaceous.

Only a few results have been reported in inland areas between the coastal areas and the Sichuan basin in the SCB, e.g., the Yichang [35] and Hengyang basins [36]. Because the poles at Ganzhou, Hengyang [36], Yichang [35] and Sichuan [20] are indistinguishable at 95% confidence level, we adopt the mean of these four poles as the reference pole (73.0° N, 239.7° E, A_{95}=4.2°) for the western stable block. So 18.9±5.5° counter-clockwise rotation is revealed in the Jishui area. We consider that the rotation at Jishui was a result of activity of the Ganjiang Fault Zone (GFZ). Deng et al. [48] suggested that the GFZ, extending about 600 km in a N20° E direction with a width of around 50–120 km, is composed of a series of faults extending in the NE or NW directions, which were also demonstrated by both gravitational and geomagnetic anomalies. The GFZ shows mainly a sinistral ductile shear from the Mesozoic and Cenozoic [48,51]. Li et al. [51] suggested three stages of an NE strike–slip fault system in eastern Hunan and western Jiangxi provinces: (1) compressional sinistral strike–slip faulting in the Jurassic, (2) extensional sinistral strike–slip faulting during the Cretaceous to Tertiary, and (3) compression with a strike–slip component in the Pliocene to the present. Although Li et al. [51] proposed that the strike–slip faulting could have resulted in a clockwise rotation of the block limited by faults, our results indicate that the counter-clockwise rotation resulted from the activity of en-echelon left-lateral strike–slip faults, suggesting ball-bearing rotation mechanics (Fig. 8). The small anticline at Jishui was the consequence of slight deformation conducted by the shear faulting. Because of the consistency of late Cretaceous poles from Sichuan [20], Yichang [35], Hengyang [36] and Ganzhou across the Yangtze block from west to east, we consider the rotation at Jishui was a local deformation. The west SCB, ranging from Sichuan Province to Jiangxi Province, has behaved as a stable block since the late Cretaceous.

The late Cretaceous poles from the east, west and interior of the SCB are listed in Table 3 (Fig. 9). Three criteria were used to select the late Cretaceous poles from the SCB. First, the age is strictly late Cretaceous. Second, the sampling areas were not subject to local deformation (rotation or displacement) under the influence of collision and penetration by the Indian

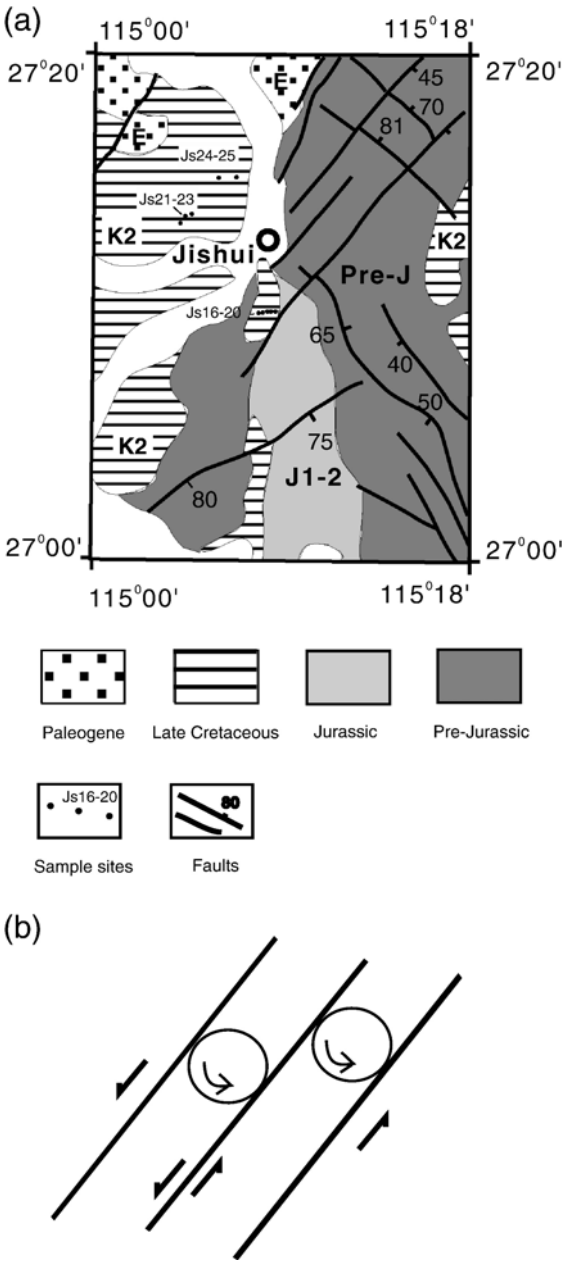


Fig. 8. (a) Simplified geological map around Jishui (modified from 1:500,000 geological maps in [38]). (b) Simple model of rotation at Jishui area.

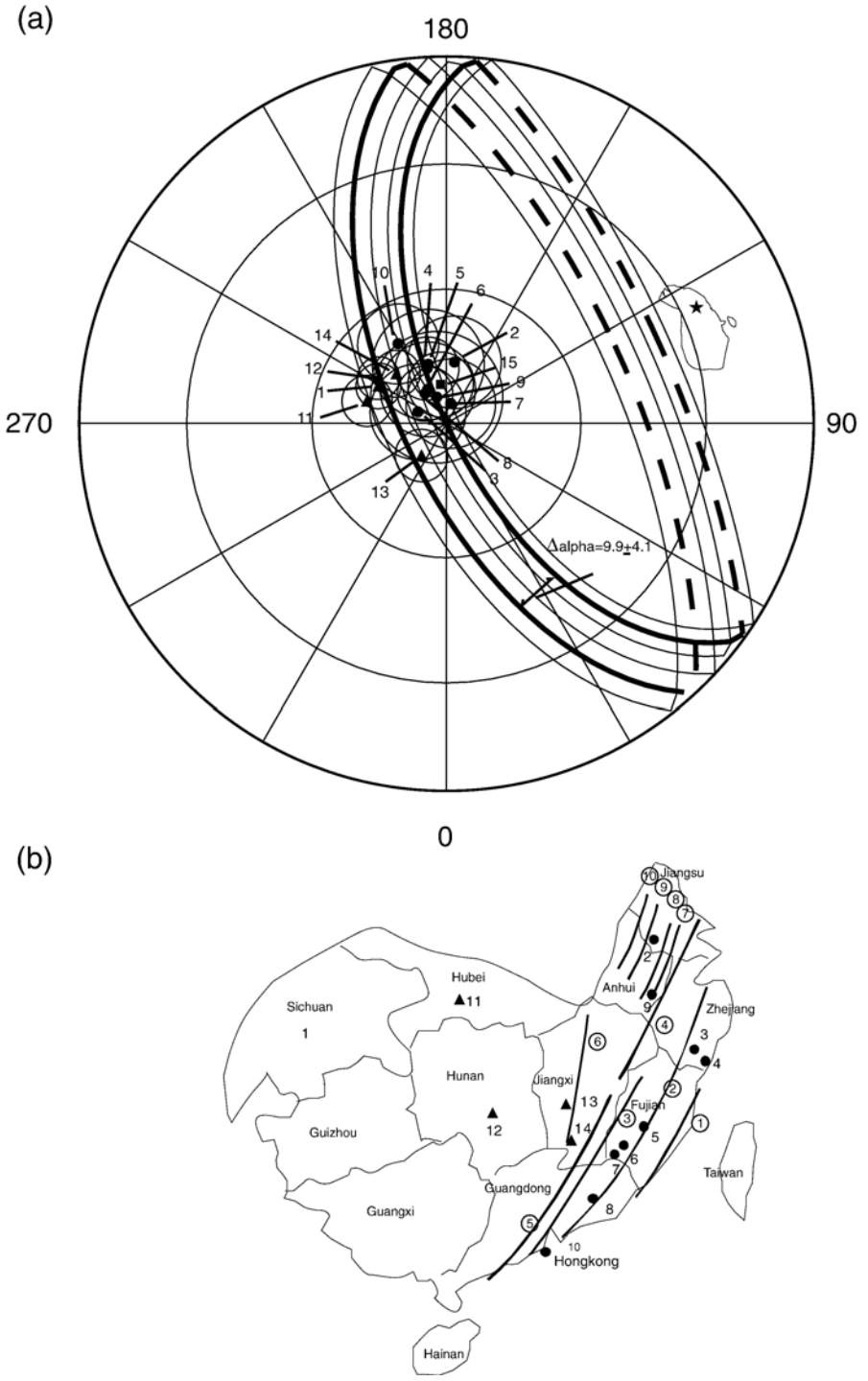


Fig. 9. (a) Equal-area projections of Late Cretaceous paleopoles from the SCB with circles of 95% confidence. Star denotes the sampling site at Ganzhou. Solid triangles are poles from inland while solid circles are poles from coastal provinces, respectively. The square is a Eurasian reference pole at 75 Ma [52]. (b) Simplified map showing the sites of available studies in South China. Solid lines are shearing fault zone in [5]: ①Changle–Nanao fault zone ②Lishui–Haifong fault zone ③ Shaowu–Heyuan fault zone ④ Tienmushan–Baijishan fault ⑤ Xingfong–Enping fault zone ⑥ Ganjiang fault zone ⑦ Guangde–Jixi fault zone ⑧ Maoshan fault zone ⑨ Fangshan–Xiaodanyang fault ⑩ Along Lower Yangtze River fault. The number of poles is the same with Table 3.

plate. Third, the number of sampling sites is more than four. It's obvious that these coeval poles do not cluster (Fig. 9). Two groups can be separated and generally related to their sampling locations situated in the coastal areas and western inland areas. Considering that the laminated samples showed clearly inclination shallowing, only data obtained from massive samples from Yichang are included for discussion. Group A (poles 1, 11–14 in Fig. 9) shows a far-side distribution while group B (2–10 in Fig. 9) displays a near-side feature. Two small circles (with radiuses of 74.0°±3.8°; and 64.1°±2.6°) centered on Ganzhou (25.9° N, 114.9° E) are fitted through these two pole groups. The different paleolatitude (ΔPlat) between these two small circles is 9.9°±4.1°, which indicates an obvious distinction in paleolatitude between the inland and coastal areas of South China. Two possible interpretations might be inferred for this distinction. First, the simplest interpretation is that the NE-oriented shearing fault system may have played an important role in the displacement of the eastern coastal block in eastern South China (Fig. 9). However, these faults are inferred as sinistral strike–slip faults that have been active since the late Cretaceous [5,48,51]. The large displacement of the block between the coast and inland SCB, ranging from Jiangsu to Guangdong provinces, is countered by the geological observations.

The second interpretation may be related to the problem of inclination shallowing in red beds. Previous studies [35,36] and our AIR results suggest the presence of inclination shallowing in west SCB, and different degrees of flattening of inclinations might occur in different basins. However, it has not been confirmed whether the inclination shallowing was present in the coastal areas. In this case, it is difficult to declare the whole SCB is a tectonic stable block.

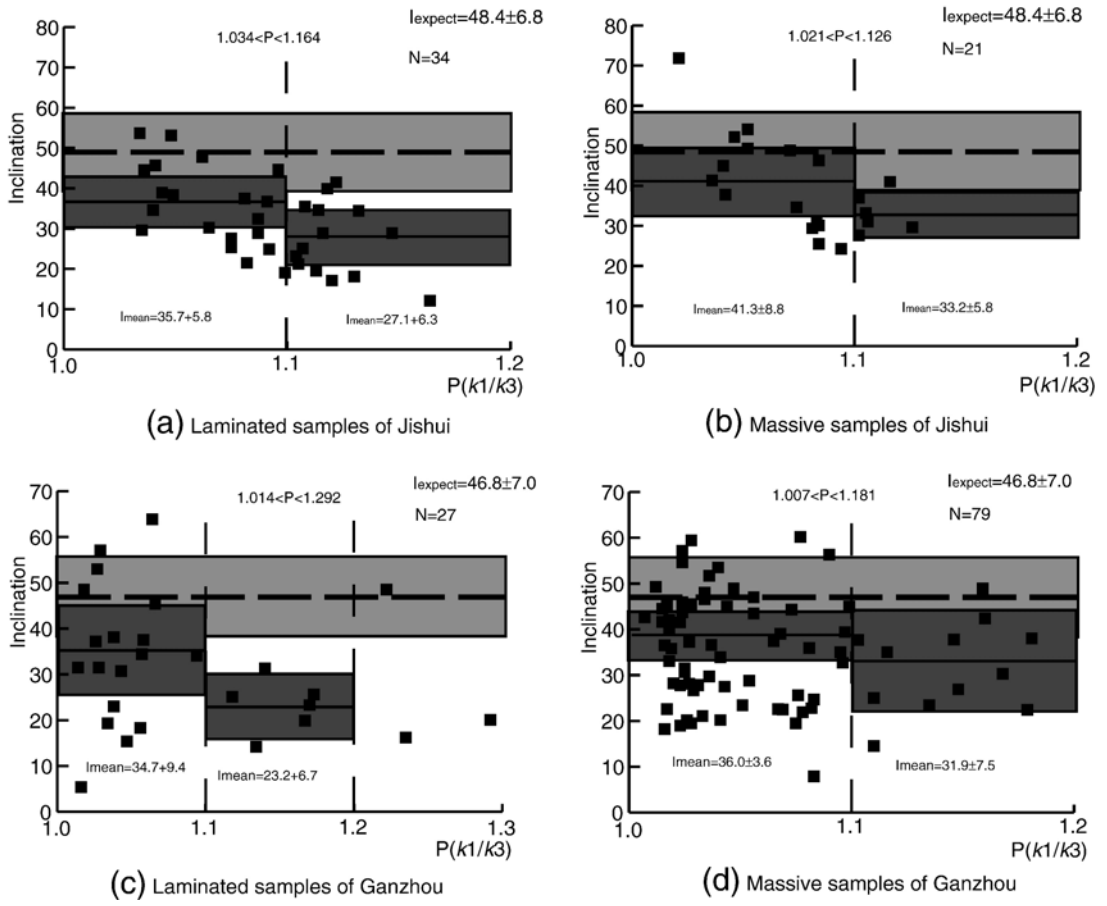


Fig. 10. Comparison between the anisotropy degree ($P=k1/k3$) and inclinations obtained from laminated and massive samples, respectively. Grey rectangles show the error of expected inclination calculated from 75 Ma reference pole, while dark grey rectangles represent the error range of mean inclination of the samples.

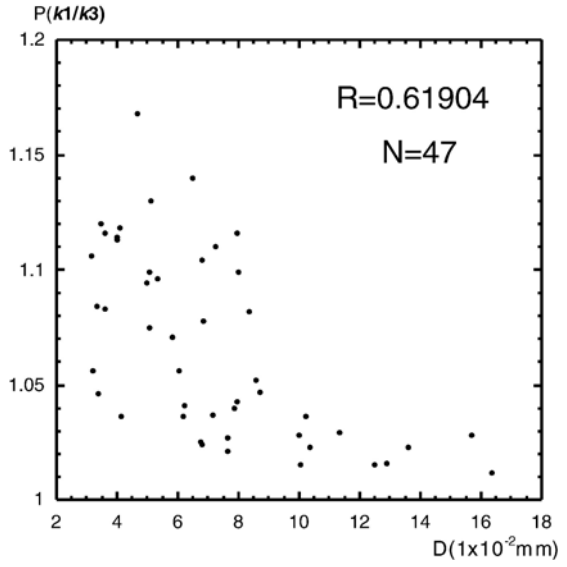


Fig. 11. Plots of anisotropy degree ($P=k1/k3$) and statistical grain sizes (D) measured from thin sections of each sample ($N=47$ samples). R is the correlative coefficient.

5.2. Inclination shallowing in South China

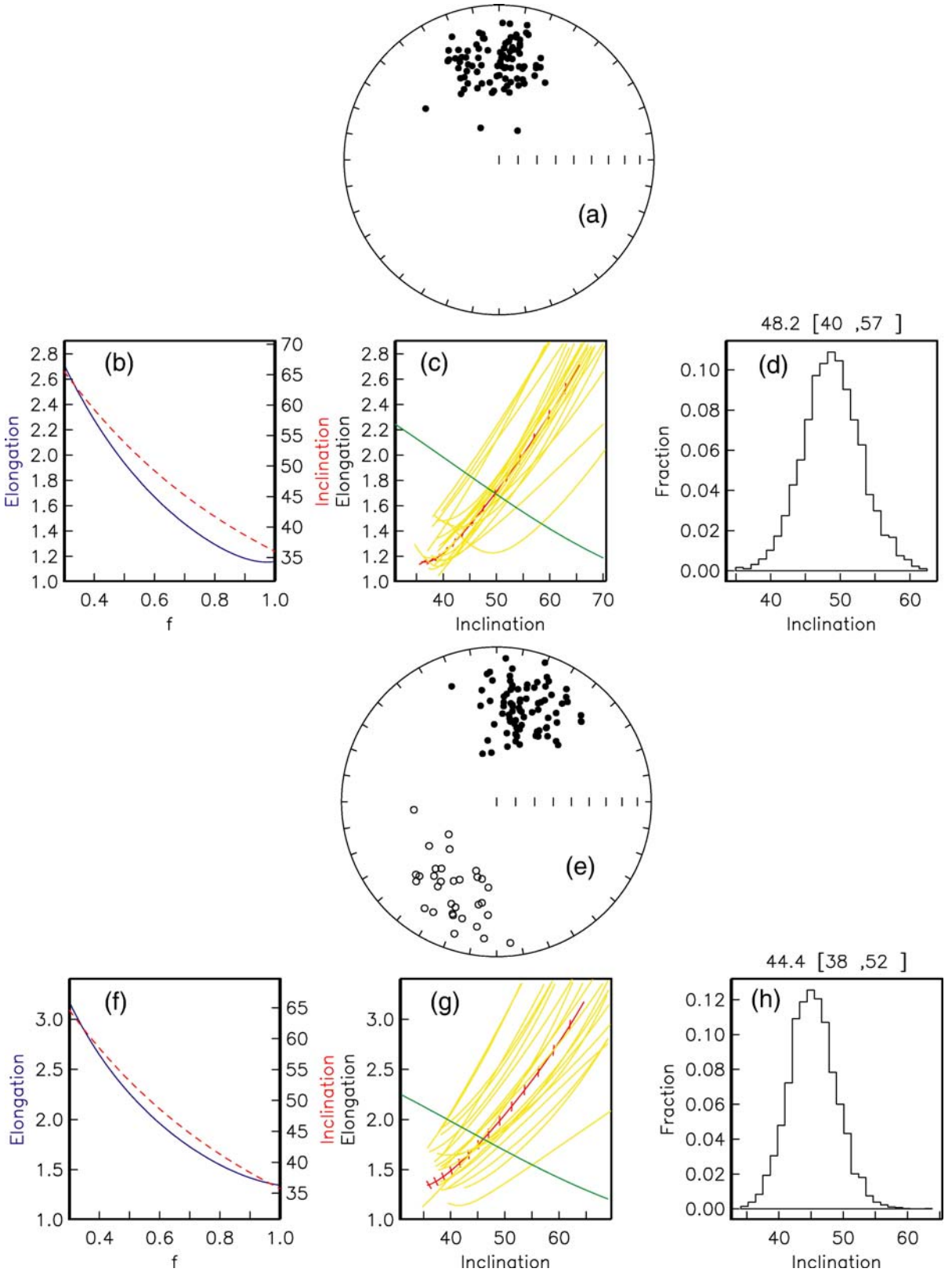
Because of a K–Ar dating age of 85.9 Ma of a thin basalt layer interbedded in red beds and the presence of the Cretaceous/Tertiary boundary at Ganzhou, we used the 75 Ma synthetic pole of Besse and Courtillot for Eurasia [52] as a reference pole for further comparison. Compared to the observed paleomagnetic directions ($I_{obs}=34.8°\pm6.3°$ at Jishui; $I_{obs}=35.6°\pm5.5°$ at Ganzhou) with the expected inclinations at Jishui (48.4°±6.8°) and Ganzhou (46.8°±7.0°), respectively, the results show significant flattening for both these areas ($\Delta I=13.6°\pm9.3°$ at Jishui, $\Delta I=11.2°\pm8.9°$ at Ganzhou). These results could imply a more than 1000 km crustal shortening between the SCB and Eurasia, such as in the Central Asian and Qinling–Dabie fold belts since the late Cretaceous, which, however, cannot be confirmed by present geological observations. The situation becomes similar to the inclination shallowing reported from Central Asia [32–34]. Although several hypotheses have been proposed to explain this inclinational flattening, such as non-dipole field or regional geomagnetic field anomaly, poorly constrained age of red beds and poorly constrained APWP for Eurasia [30–34,36], there is still a puzzle over the anomaly of the geomagnetic field during the Mesozoic and Cenozoic [35,53]. The late Cretaceous red beds of our sampling areas are well dated by paleontological and geochronological methods. Thus the syn-sedimentary or compaction effect and poorly constrained APWP may be discussed as follows:

AMS is regarded as an indicator of deposit environment, which could be closely correlated with paleomagetic inclination shallowing conducted by syn-sedimentary compaction [35,54]. Our results confirm this opinion (Fig. 10). In both sampling areas, laminated samples (34 samples at Jishui and 27 samples at Ganzhou) have higher anisotropy degrees (P) than those of massive samples (21 samples at Jishui and 79 samples at Ganzhou). The mean inclinations of higher P samples are usually flattening. However, some authors [31,36] argued that the anisotropy degree of red beds, bearing hematite as the ChRM carrier, could not be related to the degree of shallowing due to the presence of dominant diamagnetic/paramagnetic minerals, compared to minor ferromagnetic minerals. In addition, Jezek and Gilder [55] suggest that faster sedimentation rates might produce more inclination shallowing than slower rates. It is clear that faster sedimentation may conduct the coarser grains conserved, and some experiments indicate that coarse-grained samples had little compaction-caused inclination error [56]. This confusion needs more study. We thus chose two to three samples from each site, and got a total of 47 thin sections for observation. Sediment grain sizes were measured under the microscope, adopting quartz grain as the object of reference. In each thin section, 300 grains were measured to get a mean grain size. Fig. 11 shows the correlation of grain size to anisotropy degree (P). It is clear that anisotropy degree decreases while the grain sizes increase, even if the correlative coefficient $R=0.61904$ is low. The anisotropy degrees are generally lower than 1.03, while the sediments are of coarse grain sizes (mean diameter > 0.1 mm).

At the same time, inclination results from the Yichang [35], Hengyang [36] and Ganzhou areas are compared with expected inclinations calculated from the Eurasian reference pole at 75 Ma [52]. The results are listed in Table 4, which shows that the high anisotropy degree P is present in the red bed of the Hengyang basin, whereas the difference (ΔI) between expected and

Table 4
Comparison between the anisotropy degree (P) and flattening deduced from observed inclination and that of expected in the Late Cretaceous red beds in South China

Area	Anisotropy degree (P)	Difference between I_{obs} and IF (ΔI)
Yichang, Hubei	1.017–1.227	28.3±8.2
Hengyang, Hunan	1.02–1.32	17.7±8.9
Jishui, Jiangxi	1.007–1.292	13.6±9.3


(a)
(b)
Elongation
Inclination
f
(c)
Elongation
Inclination
Inclination
48.2 [40 ,57]
(d)
Fraction
Inclination
(e)
(f)
Elongation
Inclination
f
(g)
Elongation
Inclination
Inclination
44.4 [38 ,52]
(h)
Fraction
Inclination

observed inclinations is moderate. Thus, we consider that the simple bulk anisotropy degree is not a good reference parameter for detecting inclination shallowing.

Moreover, AIR results derived from the Hengyang [36] and Jishui basins reveal that a flattening inclination can be produced by deposit compaction, although the corrected inclination is still a little lower than that calculated from the Eurasian APWP. Sun et al. [36] suggests that the obviously shallowing in SCB might result from the non-rigid behavior of the huge Eurasian plate. As mentioned by Cogne [33], because of the large-size Eurasia, covering more than 180° in longitude, small rotations in declination in western Eurasia can translate into large latitudinal motions on the eastern margin, such as in eastern Asia and Siberia. The evidence that the late Cretaceous poles recently obtained from volcanic rocks in Mongolia and Siberia, also showing a much shallower inclination seems to prove this suggestion [57,58]. However, this interpretation cannot explain the obviously difference of paleolatitude between results from the coastal provinces and internal SCB. In addition, although with a large confidence limit, the late Cretaceous pole reported from dykes in Hong Kong [24] has also showed distinct higher paleolatitude than those reported from Sichuan, Hunan and Jiangxi (Fig. 9). Thus, we suggest that sedimentation and/or compaction might have been the main mechanics for the inclination shallowing. On the other hand, only small amounts of sediment were conserved in the numerous small basins distributed in the coastal areas. This could have been conducive to the low degree of sedimentary compaction and flattening of inclination.

Finally, the elongation/inclination (*E*/*I*) correction model of Tauxe and Kent [37] was used to evaluate the inclination error from the Jishui and Ganzhou areas. The results are shown in Fig. 12, in which the corrected inclinations of Jishui and Ganzhou are $48.2^{\circ}{}^{57}_{40}$ and $44.4^{\circ}{}^{52}_{38}$, respectively. These two paleo-inclinations are not much different from those calculated from the 75 Ma pole for Eurasia (48.4°±6.8° at Jishui, 46.8°±7.0° at Ganzhou), which may imply that the synthetic apparent polar wander path for Eurasia could be used to constrain the tectonic evolution of SCB. Thus, we consider that the syn-sedimentary or compaction resulted in obviously inclination shallowing in the late Cretaceous red beds in South China.

6. Conclusion

We have reported new late Cretaceous paleomagnetic results from the Jishui and Ganzhou basins in the SCB. The ChRM directions from the late Cretaceous red beds passed both fold and reversal tests, which indicates a primary magnetization of these late Cretaceous red beds. However, an obvious inclination shallowing is inferred by AIR measurement and *E*/*I* model of Tauxe and Kent [37]. We agree that the inclination shallowing conducted by sedimentation and/or compaction was the main mechanism in SCB. However, the question of the presence or otherwise of shallowing in red beds across the basin and ranges in South China, e.g. the coastal provinces, needs further study.

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Fig. 12. Equal area projection of paleomagnetic directions of samples from Jishui (a) and Ganzhou (e) areas. Left: plots of elongation and inclination versus different values of *f* for Jishui (b) and Ganzhou (f) areas. Middle: elongation versus inclination plots for TK03.GAD model and for the paleomagnetic data (barbed line) for different values of *f* for Jishui (c) and Ganzhou areas (g). The crossing points represent the inclination/elongation pair most consistent with the TK03.GAD. Right: histograms representing crossing points from bootstrapped datasets of Jishui (d) and Ganzhou areas (h).

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